

YELP : Reviews & Sentiment Analysis

Create tables for json objects

1. create or replace table yelp_review(
 business VARIANT
);
2. create or replace table yelp_business(
 business VARIANT
);

Get data from AWS-S3 to Snowflake tables

1. copy into yelp_review
 from 's3://bucket/path'
 credentials = (
 AWS_KEY_ID = 'your-access-key-id'
 AWS_SECRET_KEY = 'your-secret-key'
)
 file_format = (type = JSON);
2. copy into yelp_business
 from 's3://bucket/path'
 credentials = (
 AWS_KEY_ID = 'your-access-key-id'
 AWS_SECRET_KEY = 'your-secret-key'
)
 file_format = (type = JSON);

Create Python UDF for Sentiment analysis

```
create or replace function analyze_sentiment(text STRING)
returns STRING
language PYTHON
runtime_version = '3.10'
packages = ('textblob')
handler = 'sentiment_analyzer'
as $$
from textblob import TextBlob
def sentiment_analyzer(text):
```

```

analysis = TextBlob(text)
polarity = analysis.sentiment.polarity

if polarity > 0:
    return 'Positive'
elif polarity == 0:
    return 'Neutral'
else:
    return 'Negative'
$$;

```

Get Sentiment of few reviews

```

select
    review:text::STRING as review_text,
    analyze_sentiment(review:text::STRING) as sentiment
from yelp_review
limit 5
offset 10000;

```

	REVIEW_TEXT	SENTIMENT
1	best onion rings ever. best burgers ever. best everything. well.. besides the wait but that's just som	Positive
2	Well stocked store and very helpful. Robin does a good job of cleaning our pool.	Positive
3	Delicious, fresh food, great wine selection, creative seasonal cocktails and great service! Beautiful	Positive
4	WORST PLACE TO LIVE!!! The management sucks, nothing ever gets done & they are always trying	Negative
5	before you get too confused, take a breath and remember that stepford and the truman show are v	Positive

Unpack JSON as table

1. create table tbl_yelp_biz as

```

select
    business:business_id::string as business_id,
    business:city::string as city,
    business:state::string as state,
    business:review_count::string as review_count,
    business:stars::number as stars,
    business:categories::string as categories
from yelp_business;

```

```
select *
from tbl_yelp_biz
limit 5;
```

	<u>A</u> BUSINESS_ID	<u>A</u> CITY	<u>A</u> STATE	<u>A</u> REVIEW_COUNT	# STARS	<u>A</u> CATEGORIES
1	4GHLneyW3oxaocy2udlbHA	Edmonton	AB	6	5	Indian, Restaurants
2	L1DN799Mg7DS3Xfsl3FHKQ	Mt. Juliet	TN	6	5	Caterers, Food, Barbeque, S
3	dTwcyPcEm63U3RVK2_MvmA	Ballwin	MO	80	5	Beaches, Hiking, Boating, Fis
4	EjvezvrSdxb7XbluUd7KDg	Reno	NV	5	4	Massage, Beauty & Spas
5	pCz3D697nyCqHruCjJiHCA	Tucson	AZ	8	5	Junk Removal & Hauling, Loc

- create table tbl_review as
select
review:business_id::string as business_id,
review:date::date as review_date,
review:user_id::string as user_id,
review:stars::number as stars,
review:text::string as review_text,
analyze_sentiment(review) as sentiments
from yelp_review;

```
select *
from tbl_review
limit 5;
```

	<u>A</u> BUSINESS_ID	<u>REVIEW_DATE</u>	<u>A</u> USER_ID	# STARS	<u>A</u> REVIEW_TEXT
1	tc_nZBQ46584iOXQgdIBtg	2011-05-13	ph0l8s6itfqveoU0oToAFg	4	Great authentic mexican food. Try the asada tacc
2	x84mgmQenEyoOXZLaMksCA	2020-01-19	qj25oAghHiwggIVXWTci2g	1	Terrible customer service and not only was a bon
3	vwr djcP5pAhpANbSPEzAmw	2021-01-12	3PN5PNdl8ToMztFEXg26Bg	5	I love the Grillshack. My family and I order take ou
4	FdoqtMadj4ybDAL7S-yUBQ	2020-08-25	ehMink094me_5YpG3S5qoQ	5	We stop here every time we head through the are
5	BCxLbqymC1TXn2FE1BYc1A	2019-08-16	mrUb8ToOgix56KJbOlbRZQ	1	Ive been coming here for years. I used to LOVE th